Electric car: launch the design

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Plan a safe team workflow and identify the car's energy pathway.

SUCCESS CRITERIA - I can...

1. Label the battery, switch, motor and connections.
2. Identify safety checkpoints before tools or power.
3. Connect a design choice with a measurable outcome.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use "prototype" and "constraint" to explain your answer. Sentence starters are optional.

prototype: a version built to test a design

constraint: a requirement or limit that the design must meet

evidence: observations or measurements that support an idea

01 Warm-up: bring back an earlier idea

A stationary coil has constant magnetic flux **0.003 Wb** through each turn. Does that constant flux produce sustained induced emf? Explain.

Word help: flux = a measure of magnetic field passing through a chosen area

Sentence starter: I think … . I can point to …, which shows …, so … .

Marzano check - circle: ① ② ③

02 Worked example: explain the prediction

Reference: average speed = distance/elapsed time.
Use your own design and measurements. Battery → motor → drivetrain. Follow teacher training for tools and hot materials.

Worked reasoning: You can test a statement about a design if it names something you can measure. "The car is fast" can become "The car travels **3 m** in under **2 s** on this level track." This requires an average speed above **1.5 m/s**.

Your turn: Draw a test track with start, finish, distance and timing method labeled.

Sentence starter: In my work, I used … to show … . I labeled it … .

Start Finish



Distance: ____ Timing method: _____

Marzano check - circle: ① ② ③

03 Read and explain the diagram or equation

1. **Sketch your proposed car**.
2. **Label the battery**, switch, motor, driven axle and wheels.
3. **Add a theme** that will not obstruct moving parts.

Sentence starter: In my work, I used … to show … . I labeled it … .



Top view: the large rectangle is the chassis; small rectangles are wheels; the two crossbars are axles.

Marzano check - circle: ① ② ③

Electric car: launch the design

antocciphysics

Continue down each column.

Sentence starters are optional. You may plan in any language.

04 Apply the physics

Set up your build team: **choose a build lead**, safety checker and recorder. **Complete the role table**. **Name the teacher checkpoints required** before cutting, drilling, soldering and the first powered test. Leave approvals blank until they happen.

Sentence starter: My design includes … . It does what is needed by … .

Role	Team member	Approval / checkpoint
Build lead	____	____
Safety checker	____	____
Recorder	____	____

Marzano check - circle: ① ② ③

05 Explain your reasoning

1. **Write one measurable design goal** and one requirement or limit.
2. **Explain how you** will measure the goal instead of judging only by appearance.

Sentence starter: My design includes … . It does what is needed by … .

Marzano check - circle: ① ② ③

06 Review: connect an earlier idea

An ideal series loop has supply **24 V**. One component drops **4 V**. **Find the other component's drop**. **Draw the stated circuit** and **label quantities relevant** to the question.

Word help: series = connected along one unbranched current path

Sentence starter: I found … [unit]. This value means … in this situation.

Draw the complete circuit and label shared nodes.

Marzano check - circle: ① ② ③

07 Exit check

Complete: The part that converts electrical energy into rotation is ____. Before a powered test, I will check ____.

Sentence starter: I found … [unit]. This value means … in this situation.

Check / criterion	Measurement	Decision rule
____	____	____
____	____	____

Marzano check - circle: ① ② ③

08 STRETCH: test the prediction

1. Propose a second success criterion that could conflict with speed, such as straight tracking or reliability.
2. **Describe how both would be** measured.

Sentence starter: I notice … . This supports … because … .

Check / criterion	Measurement	Decision rule
____	____	____
____	____	____

Marzano check - circle: ① ② ③